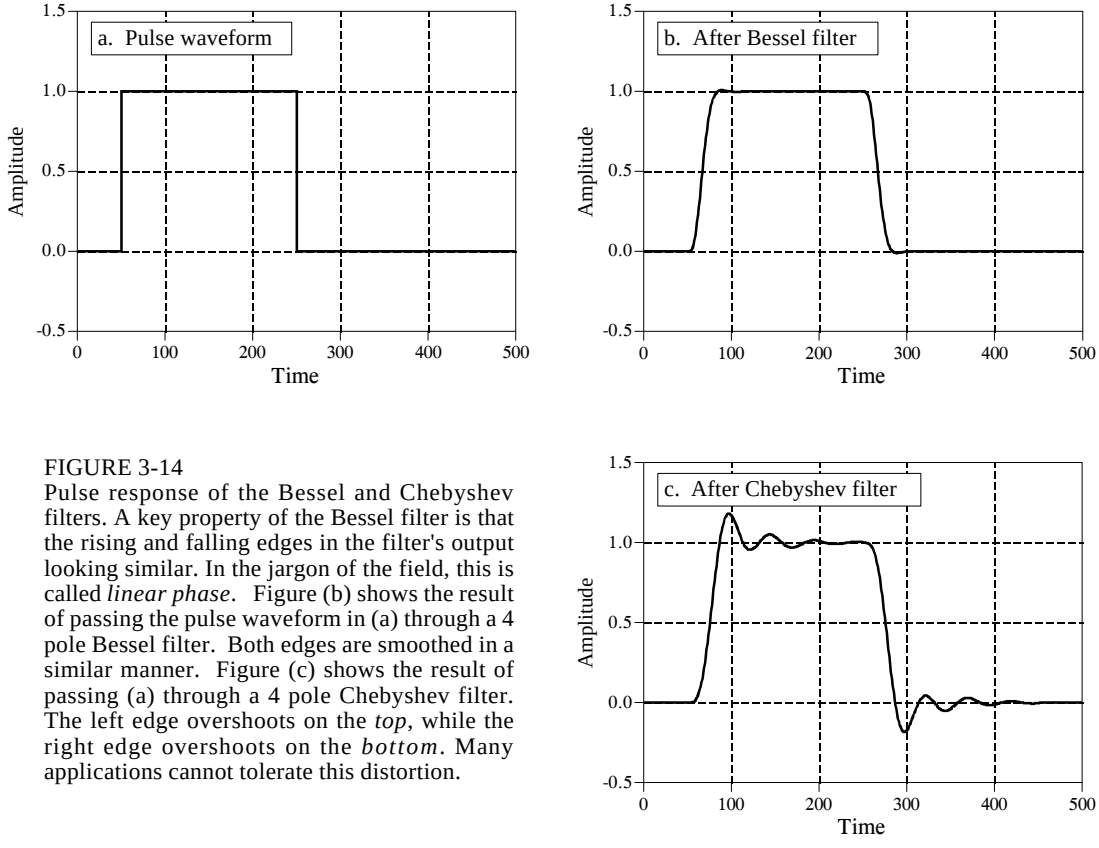

FIGURE 3-14
Pulse response of the Bessel and Chebyshev filters. A key property of the Bessel filter is that the rising and falling edges in the filter's output looking similar. In the jargon of the field, this is called *linear phase*. Figure (b) shows the result of passing the pulse waveform in (a) through a 4 pole Bessel filter. Both edges are smoothed in a similar manner. Figure (c) shows the result of passing (a) through a 4 pole Chebyshev filter. The left edge overshoots on the *top*, while the right edge overshoots on the *bottom*. Many applications cannot tolerate this distortion.

Figure 3-14 further illustrates this very favorable characteristic of the Bessel filter. Figure (a) shows a pulse waveform, which can be viewed as a rising step followed by a falling step. Figures (b) and (c) show how this waveform would appear after Bessel and Chebyshev filters, respectively. If this were a video signal, for instance, the distortion introduced by the Chebyshev filter would be devastating! The overshoot would change the brightness of the *edges* of objects compared to their *centers*. Worse yet, the left side of objects would look bright, while the right side of objects would look dark. Many applications cannot tolerate poor performance in the step response. This is where the Bessel filter shines; no overshoot and symmetrical edges.

Selecting The Antialias Filter

Table 3-2 summarizes the characteristics of these three filters, showing how each optimizes a particular parameter at the expense of everything else. The Chebyshev optimizes the *roll-off*, the Butterworth optimizes the *passband flatness*, and the Bessel optimizes the *step response*.

The selection of the antialias filter depends almost entirely on one issue: *how information is represented in the signals you intend to process*. While